

TẦNG 1. PRESENTATION LAYER

```
class AppSessionState <<global_state>> {  
    + ad_closed: bool  
    + is_logged_in: bool  
    + user_role: String  
    + username: String  
    + user_obj: UserData  
    + payment_step: bool  
    + current_page: String  
    + selected_movie: String  
    + selected_seats: List<String>  
    + config_slider: List<Any>  
    + config_list: List<Any>  
}
```

```
class MainMenu <<boundary>> {  
    + render_login_tab(): void  
    + render_register_tab(): void  
}
```

```
class AdminUI <<page>> {  
    + view_dashboard_metrics(): void  
    + manage_movie_tab(): void  
    + manage_showtime_tab(): void  
    + manage_room_tab(): void  
    + config_display_tab(): void  
    + view_top_revenue_tab(): void  
    + sell_ticket_offline_tab(): void  
    + manage_tickets_tab(): void  
}
```

```
class CustomerUI <<page>> {  
    + target_date: String  
    + target_time: String  
    + generated_ticket_ids: List<String>  
    + payment_start_time: double  
  
    + view_home_page(): void  
    + view_booking_page(): void  
    + view_history_page(): void  
}
```

```
// Hàm phụ trợ (ui_components.py)
+ navigate_to(page: String, movie: String = ""): void
+ show_advertisement(): void
+ show_popcorn_effect(): void
+ create_premium_movie_card(col: Any, title: String, genre: String, duration: int, price:
double, img_url: String): void
```

TẦNG 2. BUSINESS LOGIC LAYER

```
class AuthController {
    - _io_handler: FileIOHandler
    - _user_table: UserHashTable
    - _current_user: UserData
    - _auth_lock: Lock

    - _generate_user_id(): String
    + login(username: String, pass: String): String
    + register(username: String, pass: String, confirm_password: String): bool
    + logout(): void
    + get_current_user(): UserData
    + is_logged_in(): bool
    + is_admin(): bool
    + get_user_table(): UserHashTable
    + get_all_users(): List<UserData>
}
```

```
class AdminController {
    - _movie_controller: MovieController
    - _booking_controller: BookingController
    - _showtime_controller: ShowtimeController
    - _room_controller: RoomController

    + calculate_revenue(): double
    + count_movies(): int
    + count_tickets(): int
    + get_top_movies_by_revenue(limit: int = 10): List<MovieData>
    + sell_ticket_at_counter(movie: MovieData, showtime: Showtime, row: int, col: int):
bool/String
    + generate_report(): Map<String, Any>
    + get_active_tickets(): List<TicketData>
    + admin_delete_movie(movie_id: String): bool
    + admin_delete_room(room_id: String): bool
}
```

```

class MovieController {
    - _io_handler: FileIOHandler
    - _movie_list: MovieLinkedList

    + generate_movie_id(): String
    + add_movie(movie: MovieData): bool
    + update_movie(movie_id: String, title: String, genre: String, duration: int, description:
String, base_price: float, poster_path: String): bool
    + delete_movie(movie_id: String): bool
    + search_by_id(movie_id: String): MovieNode
    + search_by_title(title: String): MovieNode
    + get_movie_list(): MovieLinkedList
    + get_movie_data(): List<MovieData>
    + movie_exists(movie_id: String): bool
    + save_data(): void
    + load_ui_config(): Map<String, List<String>>
    + save_ui_config(slider_titles: List<String>, list_titles: List<String>): void
}

```

```

class BookingController {
    - _io_handler: FileIOHandler
    - _showtime_controller: ShowtimeController
    - _movie_controller: MovieController
    - _room_controller: RoomController
    - _ticket_list: TicketLinkedList
    - _booking_lock: Lock

    + parse_seat_id(seat_id: String): Pair<int, int>
    - _generate_ticket_id(): String
    - _sync_matrix_from_tickets(): void
    + process_booking(user: UserData, movie: MovieData, showtime: Showtime, seats:
List<Pair<int,int>>): List<String>
    + confirm_bookings_bulk(ticket_ids: List<String>): bool
    + process_counter_booking(movie: MovieData, showtime: Showtime, row: int, col: int):
bool/String
    + get_booking_history(user_id: String): List<TicketData>
    + find_ticket(ticket_id: String): TicketNode
    + generate_ticket_info(ticket_id: String): String
    + get_ticket_list(): TicketLinkedList
    + get_ticket_data(): List<TicketData>
    + cleanup_unfinished_reservations(timeout_minutes: int = 5): void
    + admin_cancel_ticket(ticket_id: String): bool
    + refresh_booking_data(): void
}

```

```
}
```

```
class ShowtimeController {
```

```
- _io_handler: FileIOHandler  
- _movie_controller: MovieController  
- _showtime_list: ShowtimeLinkedList
```

```
+ generate_showtime_id(): String  
+ add_showtime(st: Showtime, movie_controller: MovieController): bool  
+ update_showtime(showtime_id: String, new_start_time: String): bool  
+ delete_showtime(showtime_id: String, ticket_controller: Any = None): bool  
+ find_showtime(showtime_id: String): ShowtimeNode  
+ check_seat_status(showtime_id: String, row: int, col: int): bool  
+ book_seat(showtime_id: String, row: int, col: int): bool  
+ release_seat(showtime_id: String, row: int, col: int): bool  
+ get_showtime_list(): ShowtimeLinkedList  
+ get_showtime_data(): List<Showtime>  
+ change_seat_status_by_admin(showtime_id: String, row: int, col: int, new_status: int):
```

```
bool
```

```
+ get_schedule_by_date(target_date: String, movie_controller: MovieController):
```

```
List<List<Any>>
```

```
+ extract_date(showtime: Showtime): String  
+ extract_time(showtime: Showtime): String  
+ get_unique_sorted_dates(showtimes_list: List<Showtime>): List<String>  
+ get_unique_sorted_times(showtimes_list: List<Showtime>): List<String>  
+ get_showtimes_by_movie(movie_id: String): List<Showtime>  
+ find_exact_showtime(movie_id: String, target_date: String, target_time: String):
```

```
Showtime
```

```
}
```

```
class RoomController {
```

```
- _io_handler: FileIOHandler  
- _room_list: RoomLinkedList
```

```
+ add_room(room: Room): bool  
+ update_room(room_id: String, new_name: String): bool  
+ delete_room(room_id: String): bool  
+ find_room(room_id: String): RoomNode  
+ find_room_by_name(room_name: String): Room  
+ get_room_list(): RoomLinkedList  
+ get_room_data(): List<Room>  
+ count_rooms(): int
```

```
}
```

TẦNG 3. DATA STRUCTURE LAYER

```
class Node {
    + data: Any
    + next: Node
    + get_data(): Any
    + set_data(data: Any): void
    + get_next(): Node
    + set_next(next_node: Node): void
}
```

```
class UserNode <<node>> {} // Các hàm thực thể kế thừa hàm Node cha
class TicketNode <<node>> {}
class MovieNode <<node>> {}
class ShowtimeNode <<node>> {}
class RoomNode <<node>> {}
```

```
class Array2D {
    + rows: int
    + cols: int
    + data: List<Any>
    + get_val(r: int, c: int): Any
    + set_val(r: int, c: int, value: Any): void
}
```

```
class UserHashTable {
    - _SIZE: int
    - _table: List<UserNode>
    + hash_function(key: String): int
    + insert(username: String, value: UserData): void
    + get(username: String): UserData
    + contains(username: String): bool
    + remove(username: String): bool
    + display(): void
    + get_table(): List<UserNode>
    + get_all(): List<UserData>
}
```

```
class TicketLinkedList {
    - _head: TicketNode
    - _tail: TicketNode
    + add_ticket(ticket: TicketData): void
    + find_ticket(ticket_id: String): TicketNode
}
```

```

+ remove_ticket(ticket_id: String): bool
+ find_by_user(user_id: String): List<TicketData>
+ traverse(): void
+ get_head(): TicketNode
+ get_tail(): TicketNode
+ clear(): void
}

```

```

class MovieLinkedList {
- _head: MovieNode
- _tail: MovieNode
+ add_movie(movie: MovieData): void
+ search_movie(title: String): MovieNode
+ search_id(movie_id: String): MovieNode
+ remove_movie(movie_id: String): bool
+ sort_by_revenue_logic(): void
+ traverse(): void
+ get_head(): MovieNode
}

```

```

class ShowtimeLinkedList {
- _head: ShowtimeNode
- _tail: ShowtimeNode
+ add_showtime(st: Showtime): void
+ find_showtime(showtime_id: String): ShowtimeNode
+ find_by_movie(movie_id: String): List<Showtime>
+ remove_showtime(showtime_id: String): bool
+ traverse(): void
+ get_head(): ShowtimeNode
}

```

```

class RoomLinkedList {
- _head: RoomNode
- _tail: RoomNode
+ add_room(room: Room): void
+ find_room(room_id: String): RoomNode
+ remove_room(room_id: String): bool
+ traverse(): void
+ get_head(): RoomNode
}

```

```

// Môi quan hệ kế thừa
UserCode --> Node
TicketNode --> Node

```

MovieNode --> Node
ShowtimeNode --> Node
RoomNode --> Node

TẦNG 4. DATA MODEL LAYER

```
class SeatStatus <<enum>> {  
    + EMPTY: int = 0  
    + RESERVED: int = 1  
    + BOOKED: int = 2  
    + BLOCKED: int = -1  
}
```

```
class UserData <<entity>> {  
    - _username: String  
    - _password: String  
    - _role: String  
    - _user_id: String  
  
    + get_username(): String  
    + get_password(): String  
    + get_role(): String  
    + get_user_id(): String  
    + check_password(password: String): bool  
}
```

```
class TicketData <<entity>> {  
    - _ticket_id: String  
    - _user_id: String  
    - _movie_id: String  
    - _seat_id: String  
    - _status: String  
    - _showtime_id: String  
    - _room_id: String  
    - _price: double  
    - _booking_time: String  
  
    + get_ticket_id(): String  
    + get_user_id(): String  
    + get_movie_id(): String  
    + get_seat_id(): String  
    + get_status(): String  
    + get_showtime_id(): String
```

```
+ get_room_id(): String
+ get_price(): double
+ get_booking_time(): String
+ set_status(status: String): void
}
```

```
class MovieData <<entity>> {
- _movie_id: String
- _title: String
- _genre: String
- _duration: int
- _description: String
- _base_price: double
- _poster_path: String
- _revenue: double

+ get_movie_id(): String
+ get_title(): String
+ get_genre(): String
+ get_duration(): int
+ get_description(): String
+ get_base_price(): double
+ get_poster_path(): String
+ get_revenue(): double

+ set_title(title: String): void
+ set_genre(genre: String): void
+ set_duration(duration: int): void
+ set_description(description: String): void
+ set_base_price(base_price: double): void
+ set_poster_path(poster_path: String): void
+ set_revenue(revenue: double): void
+ add_revenue(amount: double): void
}
```

```
class SeatMatrix {
- _rows: int
- _cols: int
- _seats: Array2D

+ get_rows(): int
+ get_cols(): int
+ generate_seat_id(r: int, c: int): String
+ check_status(r: int, c: int): int
}
```

```

+ reserve_seat(r: int, c: int): bool
+ book_seat(r: int, c: int): bool
+ release_seat(r: int, c: int): void
+ get_seats(): Array2D
+ load_matrix(seats_array: List<List<int>>): void
+ set_seat_status(r: int, c: int, status: int): bool
+ reset_all(): void
}

```

```

class Room <<entity>> {
- _room_id: String
- _room_name: String
- _rows: int
- _cols: int
- _capacity: int

+ get_room_id(): String
+ get_room_name(): String
+ get_rows(): int
+ get_cols(): int
+ get_capacity(): int
+ set_room_name(new_name: String): void
}

```

```

class Showtime <<entity>> {
- _showtime_id: String
- _movie_id: String
- _start_time: String
- _room_id: String
- _seat_matrix: SeatMatrix

+ get_showtime_id(): String
+ get_movie_id(): String
+ get_start_time(): String
+ get_room_id(): String
+ get_seat_matrix(): SeatMatrix
+ set_start_time(new_time: String): void
+ get_available_seats(): int
}

```

```

class FileIOHandler {
+ base_path: String
+ users_file: String
+ movies_file: String
}

```

```
+ rooms_file: String
+ showtimes_file: String
+ tickets_file: String

+ load_users(table: UserHashTable): void
+ save_users(table: UserHashTable): void
+ load_ui_config(): Map<String, List<String>>
+ save_ui_config(slider_titles: List<String>, list_titles: List<String>): void
+ load_movies(movie_list: MovieLinkedList): void
+ save_movies(movie_list: MovieLinkedList): void
+ load_rooms(room_list: RoomLinkedList): void
+ save_rooms(room_list: RoomLinkedList): void
+ load_showtimes(showtime_list: ShowtimeLinkedList): void
+ save_showtimes(showtime_list: ShowtimeLinkedList): void
+ load_tickets(ticket_list: TicketLinkedList): void
+ save_tickets(ticket_list: TicketLinkedList): void
+ loadData(users: UserHashTable, movies: MovieLinkedList, rooms: RoomLinkedList,
showtimes: ShowtimeLinkedList, tickets: TicketLinkedList): void
  + saveData(users: UserHashTable, movies: MovieLinkedList, rooms: RoomLinkedList,
showtimes: ShowtimeLinkedList, tickets: TicketLinkedList): void
}
```